MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8252W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, pulse modulator

(A) See figure 1

Mounting position: Any

Weight: 12 ounces nominal (340.2 grams)

ABSOLUTE RATINGS: Pulse modulation

Parameter: / Unit:	Ef V	Eb kVdc	Ec1 Vdc	ec1 v	Ec2 kVdc	Pp W	Pg2 W	Pg1 W	tk sec	ib a (see note 1)	Seal temp (glass) °C	Seal temp (ceramic) °C
Maximum:	28.6	20	-1,000	300	1.5	60	8	1	---	18	200	250
Minimum:	23.4	---	---	---	---	---	---	---	180	---	---	---
TEST CONDITIONS:	25.0	1.2	---	---	1.0	---	---	---	180	---	(see note 11)	(see note 11)

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission	6	ec1 = ec2 = eb = 1,000 v; tk = 300 (min)	0.65	II	is	45	---	a
1256	Electrode current (screen)	-	Ec1/Ib = 100 mAdc; Eb = 150 Vdc; Ec2 = 100 Vdc	0.65	II	Ic2	0	5.0	mAdc
1261	Electrode voltage (1) (grid)	-	Ec1/Ib = 50 mAdc	0.65	II	Ec1	-190	-290	Vdc
1261	Electrode voltage (2) (grid)	-	Ec1/Ib = 1.0 mAdc	0.65	II	Eco	---	-350	Vdc
1266	Total grid current	2,6	Ec1/Ib = 50 mAdc	0.65	II	Ic1	---	-20	µAdc
1266	Primary grid emission (control)	10	Ic1 = 65 mAdc; t = 15 sec; anode and screen grid floating	0.65	II	Isg1	---	-50	µAdc
1266	Primary grid emission (screen)	10	Ec1 = 0; t = 15 sec; Ic2 = 100 mAdc; anode floating	0.65	II	Isg2	---	-150	µAdc
1356	Grid-pulse operation (1)	3,6	Ec1 = -800 Vdc; Ebb = 25 kVdc; Ec2 = 1,350 Vdc; Rt = 1,030 to 1,070 ohms; t = 300 sec (max)	0.65 0.65 0.65	II II II	iL ΔiL Ic2	20 --- Never negative	--- 10	a %
	Quality conformance inspection, part 2								
1301	Heater current	-	Ef = 27.0 V	---	---	If	1.95	2.35	A
---	Torque	4		---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode grounded	---	---	Cgp Cin Cout	--- 35 6.0	2.0 50 11	pF pF pF
1356	Grid-pulse operation (2)	3	Ec1 = -800 Vdc; Ebb = 20 kVdc; Ec2 = 1,250 Vdc; Rt = 1,050 to 1,100 ohms; t = 300 sec (max)	--- --- ---	--- --- ---	iL ΔiL Ic2	16 --- Never negative	--- 10	a %
	Quality conformance inspection, part 3								
---	Life test (1)	3,8	Group C; Ebb = 20 kVdc; Ec1 = -800 Vdc; Ec2 = 1,250 Vdc; Rt = 1,030 to 1,070 ohms; Du = 0.001; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test end points (1):								
1356	Grid-pulse operation (2)	-		---	---	iL	15	---	a
						ΔiL	---	15	%
1266	Total grid current	-		---	---	Ic1	---	-100	μAdc
1231	Peak emission	-		---	---	is	40	---	a
1266	Primary grid emission (control)	-		---	---	Isg1	---	-50	μAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-150	μAdc
---	Life test (2)	9	Group C; Ef = 27.0 V; Ec1 = -90 Vdc (min); anode and screen grid floating; t = 500 hours	---	---	---	---	---	---
---	Life-test end point (2):								
1301	Heater current	9	Ef = 27.0 V	---	---	If	1.95	2.35	A
---	Vibration	5,12	Ef = 24 V; Ebb = 250 Vdc; Rt = 2,500 ohms; Ec2 = 90 Vdc; Ec1/Ib = 5.0 mAdc; F = 10 to 200 Hz; DA = 0.08 inch, 10 to 50 Hz; accel = 10G, 50 to 200 Hz	---	---	Ep	---	1.5	Vac
---	Vibration end point:								
1356	Grid-pulse operation (2)	-		---	---	iL	16	---	a
						ΔiL	---	10	%
						Ic2	Never negative		
---	Long-duration shock	7,12	Ef = 25.0 V; no other voltages applied; shock = 50G peak (min); duration = 11±2 ms; total impacts = 18; X, Y, and Z axes	---	---	---	---	---	---
---	Long-duration shock end point:								
1356	Grid-pulse operation (2)	-		---	---	iL	16	---	a
						ΔiL	---	10	%
						Ic2	Never negative		

NOTES:

1. The drive pulse tr and tf significantly effect anode dissipation. In all cases of operation the rated element dissipations shall not be exceeded. The information contained on figure 2 may be used as a guide to safe cathode current for various combinations of pulse repetition rate and pulse duration.

2. Tube to operate with constant or decreasing grid current for 2 minutes. If the tube should operate with a rise of grid current, the grid current shall become constant or decrease within 5 minutes.

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NOTES: -Continued

3. Driver pulse duration 2 µs minimum, measured at 95 percent of maximum amplitude, with a tr less than 0.2 µs and a tf less than 0.4 µs. Variation in amplitude over 80 percent of the top portion of the pulse shall be not more than 5 percent. Pulse repetition rate shall be 300 to 500, with the exact frequency at the option of the manufacturer.

   Input signal to the control grid shall drive the grid positive by 225 ±25 volts at the top of the pulse.

   The variation of amplitude of the output pulse shall be not greater than specified.

   The tube shall operate for a 2-minute period without arcing within the tube which would kickout the overcurrent relay.

   Tube shall be tested in the circuit shown on figure 3, or equal.

   A Westinghouse type MN overcurrent relay, style 1158584 (Navy type CAY29095), or equal, which has a 50 to 200 mAdc range shall be used in conjunction with an ac control relay in grid-pulse operation testing and life testing of this tube type. The dc overcurrent relay shall be equipped with coil No. 1158882 and shall be adjusted to 200 mAdc. The ac control relay used in conjunction with the dc overcurrent relay shall open the ac line approximately 20 milliseconds after the overcurrent relay opens. A device shall be incorporated into the relay circuit which will delay the reapplication of anode voltage for a minimum period of 15 seconds after each kickout and shall count the number of kickouts.

4. The adapter wafer shall be subjected to a gradually applied torque of 12.0 pound-inches. This test shall not cause broken leads, welds, soldered joints, broken glass, or cracked glass.

5. The tube under test shall be vibrated (simple harmonic motion) with an ascending sweep only. Each tube shall be vibrated in the X, Y, and Z axes, under the specified conditions. The impedance of the anode and screen voltage supplies shall not exceed that of a 40 µF capacitor at 10 Hz. The value of the alternating voltage (Ep) produced across the resistor (Rp) shall be measured with a suitable device. This device shall have an appropriate voltage range and shall have the ability to measure, with an error of less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 5,000 Hz. Thermal or VU meters may be used provided the shunting effect on the anode load does not result in an error greater than the 10 percent allowed above. The time for covering the sweep range of 10 to 50 Hz shall be 3 minutes minimum, with the time for covering the sweep range of 10 to 200 Hz being 4 to 15 minutes. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration output voltage in each of the three axes. If at the end of 60 seconds, the vibration output is increasing, the vibration shall be continued until there is no further increase.

6. The tests listed below shall be the first tests performed after the holding period in the following order:

First:	Peak emission
Second:	Total grid current
Third:	Grid-pulse operation (1)

7. The tube under test shall be shocked in the X, Y, and Z axes, with six blows in each axis. An approved short tester shall be used to indicate any shorts between the cathode and control grid and between the control grid and screen grid. No permanent shorts and no more than one temporary short is allowable during the test. The applied shock shall be an approximate half-sine wave motion, with duration measured at the zero axis level.

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NOTES: -Continued

8. During life test, any tubes which will kick out the overcurrent relay more than four times in any 24-hour period shall be considered a failure, excluding an initial stabilization period of 15 minutes.

9. Heater voltage shall be cycled approximately 5 minutes "on" and 25 minutes "off". No grid-cathode shorts are permitted during or after life test. The specified minimum t = 500 hours includes 1,000 complete "on-off" cycles. For qualification, data on a sample of 10 tubes shall be submitted.

10. Primary-grid-emission tests shall be made in a circuit with suitable rectifiers and resistors, to limit the voltage applied to the grid under test during the heating half-cycles, while allowing the emission measurement to be made at 110 volts rms for the control grid and 750 volts rms for the screen grid during the measuring half-cycles.

11. In all cases of operation with ambient temperature above 25°C, above sea level, poor ventilation, or a combination of these factors, a heat dissipating connector is allowable on the anode terminal cap. Adequate ventilation may be provided and the maximum temperature rating for envelope and seals shall not be exceeded.

(A) 12. Test to be performed every three months, with sampling as follows:

n1 = 4	$c_1 = 0$	Where c2 represents the total allowable failures for the first and second samples combined.
n2 = 4	c2 = 1;	

In case of failure, the failing test or the failing dimension(s) shall become quality conformance inspection (QCI), part 2, AQL 6.5, inspection level S3, for three successive successful submissions, at which time the testing may revert to the quarterly basis.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES
Army - AV, MI

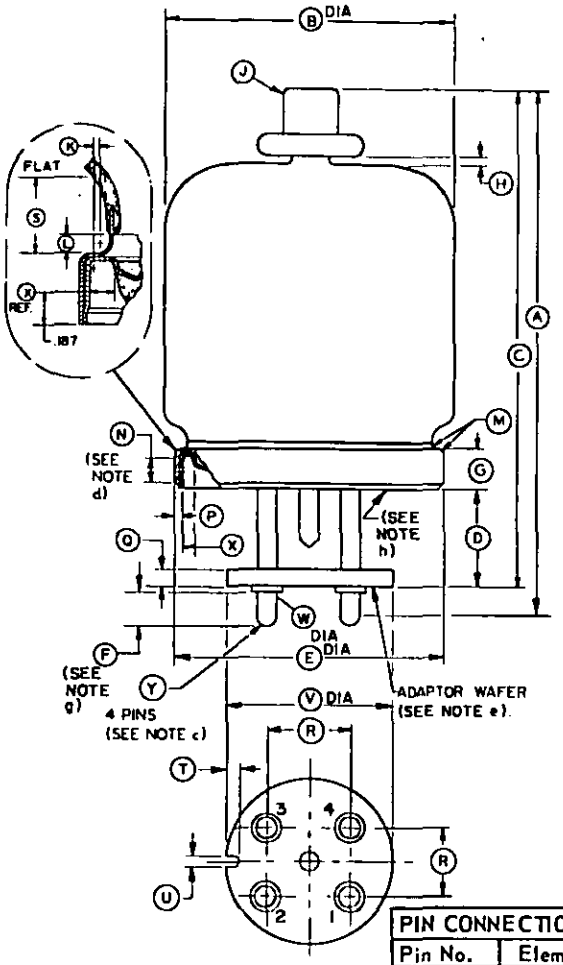
User activities:
Army - MU, AV, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5960-2844)

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PIN CONNECTIONS	
Pin No.	Element
1	h
2	g2
3	g1
4	hk
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.750 (146.05)	6.000 (152.40)
B		3.063 (77.80)
E	2.885 (73.28)	2.905 (73.79)
P	.043 (1.09)	.057 (1.45)
V	1.788 (45.42)	1.813 (46.05)
Quality conformance inspection, part 3 (see note 12)		
C	5.344 (135.74)	5.594 (142.09)
D	1.125 (28.58)	1.250 (31.75)
F	.328 (8.33)	
G	.438 (11.13)	.500 (12.70)
H	.016 (.41)	
J	Cap: C1-41	
K	.050 (1.27)	
L	.150 (3.81)	
M		.075 (1.91)
N	.250 (6.35)	
S	.340 (8.64)	
T	.171 (4.34)	.203 (5.16)
U	.171 (4.34)	.203 (5.16)
W	.183 (4.65)	.191 (4.85)
X	.157 (3.99)	
Reference dimensions (see note b)		
Q	.188 (4.78)	
R	.687 (17.45)	
Y	.031 (.79) radius	

NOTES:

a. There shall be an electrical tie between pin 4 and clamping flange.
b. Dimensions without tolerances are for information and are not required for inspection purposes.
c. Pins shall enter gage .250 (6.35 mm) thick, with four holes .214 (5.44 mm) diameter located on .687 (17.45 mm) centers and with center hole .250 (6.35 mm) diameter.
d. Defines cylindrical surface available for clamping which must not be deformed by action of clamps.
e. The adapter wafer will not be removed.
f. Dimensions optional where not specified.
g. Defines length available for contact.
(A) h. Tube base envelope material shall be approved type ceramic.

FIGURE 1. Outline drawing of electron tube type 8252W.

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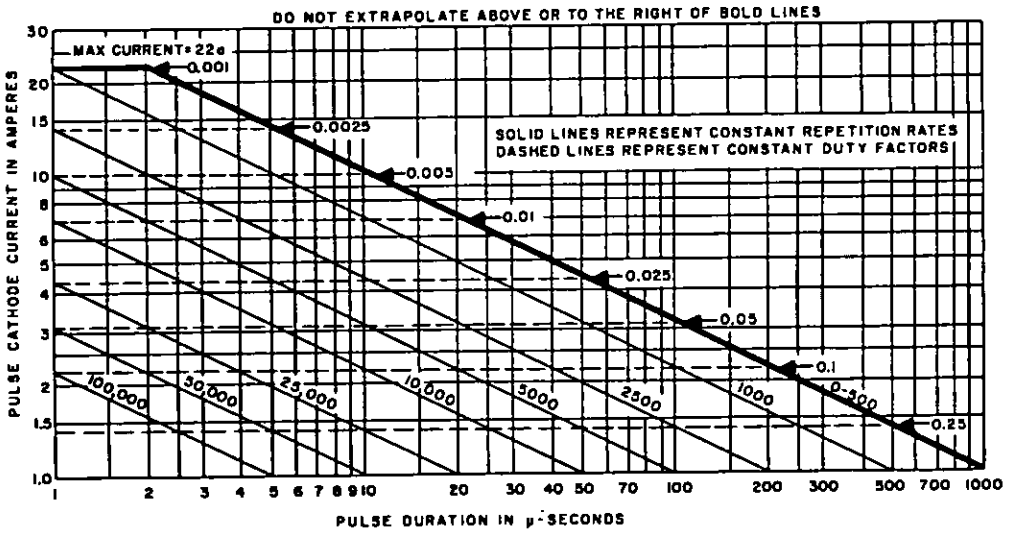


FIGURE 2. Pulse cathode current versus pulse duration (see note 1).

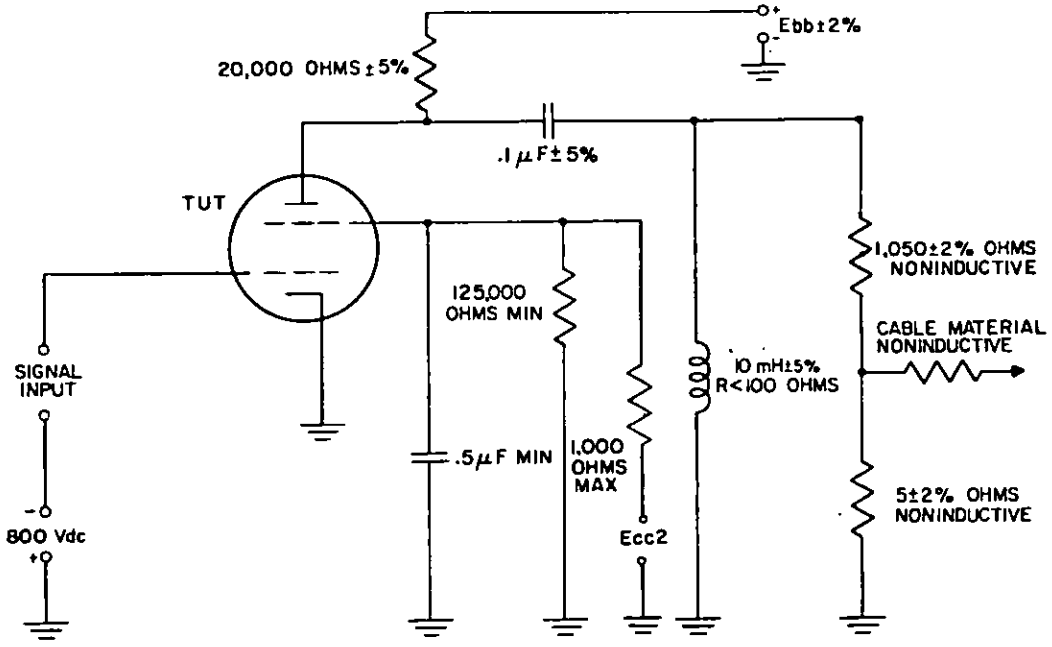


FIGURE 3. Circuit for grid pulse operation test.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1679 ELECTRON TUBE, POWER TYPE 8252W

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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